Energy-Scaled Zero-Noise Extrapolation for GKP Bosonic Qubits

Visiting Thesis Report, Westlake University

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[2] Report based on arXiv:2512.03583.

Talk Outline

Part I. Minimal GKP code

- A qubit.
- A (GKP) qubit in an oscillator hilbert space.
- Displacement stabilizers, two cosets $\longrightarrow$ two logical states
- Cavity–transmon syndrome measurement and correction.
- Finite-energy GKP states construction.

Part II. Error mitigation on a simulated GKP qubit

- Open Quantum System: Photon loss, recovery.
- Zero-noise extrapolation.
- Sweep $\bar{n}$ as the intrinsic scaling axis.
- Compare extrapolated observables and resource savings.

GKP structure $\longrightarrow$ finite energy $\longrightarrow$ ES-ZNE results

A Qubit As A Code Space

Recall: an abstract qubit is formally given by

$$V : \mathbb{C}^2 \longrightarrow \mathcal{H}_{\text{phys}}, \qquad V|\mu\rangle = |\mu_L\rangle.$$

The code space is

$$\mathcal{C} = \text{im}\, V, \qquad \dim \mathcal{C} = 2,$$

and a logical state is

$$|\psi_L\rangle = V(c_0|0\rangle + c_1|1\rangle) = c_0|0_L\rangle + c_1|1_L\rangle.$$

- **Discrete-variable:** the physical system is naturally two-level, such as a transmon or trapped ion.
- **Continuous-variable:** the physical system is a harmonic oscillator with infinitely many levels.
- *bosonic code*: Photon-number states $\longrightarrow$ Two logical states

GKP code

Gottesman, Kitaev, and Preskill introduced this bosonic code in *Encoding a qubit in an oscillator* (2001). It selects a two-level logical system from the oscillator's infinite ladder.

One Oscillator Stores One Logical Qubit

GKP code can be realized in a microwave cavity mode,

$$\mathcal{H}_{\mathrm{osc}} = \mathrm{span}\{|0\rangle, |1\rangle, |2\rangle, \ldots\}.$$

These Fock states count photons. Since

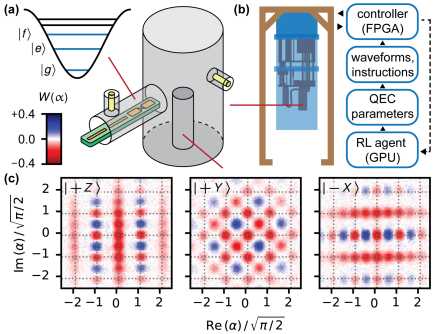
$$H_{\mathrm{osc}} = \hbar\omega_c \left(\hat{n} + \frac{1}{2}\right), \qquad \hat{n} = a^\dagger a,$$

the mean photon number is the natural dimensionless energy,

$$[a, a^\dagger] = 1, \qquad \bar{n} = \langle \hat{n} \rangle.$$

The GKP code is built from the phase-space variables

$$\hat{q} = \frac{a + a^\dagger}{\sqrt{2}}, \qquad \hat{p} = \frac{i(a^\dagger - a)}{\sqrt{2}}.$$



Displacements Define The Ideal GKP Stabilizers

The natural operation is a phase-space displacement,

$$D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a), \qquad \alpha = \frac{\Delta q + i\Delta p}{\sqrt{2}}.$$

For the square lattice we choose

$$S_q = D(\sqrt{2\pi}) = e^{-i2\sqrt{\pi}\hat{p}}, \qquad S_p = D(i\sqrt{2\pi}) = e^{+i2\sqrt{\pi}\hat{q}}, \qquad [S_q, S_p] = 0.$$

Define the code space as the simultaneous $+1$ eigenspace of these two stabilizers,

$$\mathcal{C} = \{|\psi\rangle : S_q|\psi\rangle = |\psi\rangle,\ S_p|\psi\rangle = |\psi\rangle\}.$$

Solving The Stabilizers Gives Two Cosets

The S_p condition pins support to the fine lattice,

$$S_p|q\rangle = e^{i2\sqrt{\pi}q}|q\rangle \quad \Rightarrow \quad q = m\sqrt{\pi}, \quad m \in \mathbb{Z},$$

so a code state has the form

$$|\psi\rangle = \sum_m c_m |q = m\sqrt{\pi}\rangle.$$

The S_q condition shifts by two lattice sites,

$$S_q|q\rangle = |q + 2\sqrt{\pi}\rangle, \qquad S_q|\psi\rangle = |\psi\rangle \quad \Rightarrow \quad c_{m+2} = c_m.$$

Thus

$$\Lambda = \sqrt{\pi}\mathbb{Z}, \qquad \Lambda_S = 2\sqrt{\pi}\mathbb{Z}, \qquad \Lambda/\Lambda_S \cong \mathbb{Z}_2,$$

with logical basis states

$$|0_L\rangle \propto \sum_s |q = 2s\sqrt{\pi}\rangle, \qquad |1_L\rangle \propto \sum_s |q = (2s+1)\sqrt{\pi}\rangle.$$

How The Code Is Operated

The logical state lives in the cavity, and the transmon is the ancilla.

$$H_{\mathrm{disp}} \approx \hbar\omega_r' a^\dagger a + \frac{\hbar\omega_q'}{2}\sigma_z + \hbar\chi\, a^\dagger a\, \sigma_z.$$

Pulses implement controlled displacements,

$$CD(\beta) = |g\rangle\langle g| \otimes D(-\beta/2) + |e\rangle\langle e| \otimes D(+\beta/2),$$

and a transmon readout extracts the stabilizer phase,

$$|\psi\rangle|+\rangle \longrightarrow \frac{|\psi\rangle|g\rangle + S_p|\psi\rangle|e\rangle}{\sqrt{2}}.$$

For a position displacement error,

$$S_p e^{-i\epsilon_q \hat{p}}|\psi\rangle = e^{i2\sqrt{\pi}\epsilon_q} e^{-i\epsilon_q \hat{p}}|\psi\rangle.$$

Thus the phase gives ϵ_q modulo $\sqrt{\pi}$. If

$$(\epsilon_q, \epsilon_p) \in \left[-\frac{\sqrt{\pi}}{2}, \frac{\sqrt{\pi}}{2}\right)^2,$$

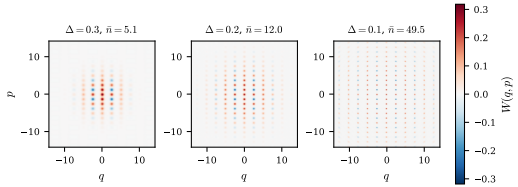
the nearest-cell correction applies the opposite displacement,

$$e^{+i\epsilon_q \hat{p}} e^{-i\epsilon_q \hat{p}}|\psi\rangle = |\psi\rangle.$$

The stabilizer cell has area 4π, two Planck cells of area 2π, matching one logical qubit.

Physical GKP Code Means Finite Energy

- In experiments: **finitely squeezed grid state** (Gaussian Peaks) from cavity–transmon control pulses.
- Squeezing: reducing quadrature uncertainty of each grid peak; **finite** squeezing leaves a residual width Δ.
- Start from the ideal comb, suppress high photon numbers, then normalize:
$$|\mu_\Delta\rangle = \mathcal{N}_\mu e^{-\Delta^2 \hat{n}} |\mu_L\rangle, \qquad \mu = 0, 1.$$
- Energy is mean photon number,
$$\bar{n}_\Delta = \langle \mu_\Delta | \hat{n} | \mu_\Delta \rangle, \qquad n_\Delta = \frac{1}{e^{2\Delta^2} - 1}.$$
- In position space this means Gaussian peaks under a Guassian envelope:
$$\psi_{\mu,\Delta}(q) \propto \sum_{s \in \mathbb{Z}} e^{-\frac{(q-(2s+\mu)\sqrt{\pi})^2}{2\Delta^2}} e^{-\frac{\Delta^2 q^2}{2}}.$$
- Smaller Δ: narrower peaks, stronger squeezing, larger $\bar{n}$.



Open-System Dynamics And Photon Loss

Open dynamics are modeled by a **Lindblad master equation**,

$$\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2}\{L_j^\dagger L_j, \rho\} \right).$$

For storage noise in the cavity, keep only photon loss:

$$H = 0, \qquad L = \sqrt{\kappa}\, a.$$

These give the **photon-loss master equation**

$$\dot{\rho} = \mathcal{D}[L]\rho = \kappa \mathcal{D}[a]\rho, \qquad \mathcal{D}[a]\rho = a\rho a^\dagger - \frac{1}{2}\{a^\dagger a, \rho\}.$$

Solving for time t defines the finite-time channel

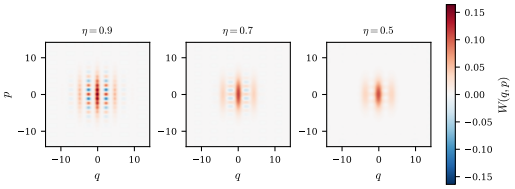
$$\rho(t) = \mathcal{N}_\eta(\rho(0)), \quad \mathcal{N}_\eta = e^{t\kappa\mathcal{D}[a]},$$

$$\eta = e^{-\kappa t}, \qquad \gamma = 1 - \eta, \qquad x = -\log \eta.$$

The same channel has the Kraus form

$$\mathcal{N}_\eta(\rho(0)) = \sum_{\ell=0}^{\infty} E_\ell \rho(0) E_\ell^\dagger,$$

$$E_\ell = \left(\frac{\gamma}{1-\gamma}\right)^{\ell/2} \frac{a^\ell}{\sqrt{\ell!}} (1-\gamma)^{\hat{n}/2}.$$

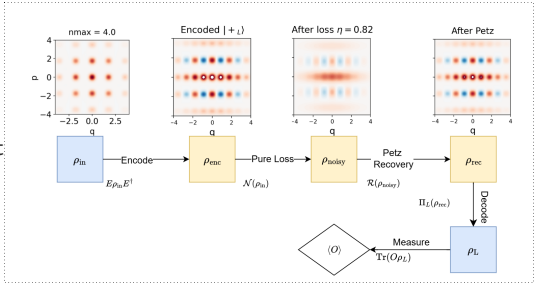


Recovery Pipeline And Logical Readout

The simulation pipeline is

$$\rho_{\mathrm{cav}} \xrightarrow{\mathcal{N}_\eta} \mathcal{N}_\eta(\rho_{\mathrm{cav}}) \xrightarrow{\mathcal{R}} \mathcal{R} \circ \mathcal{N}_\eta(\rho_{\mathrm{cav}}) \xrightarrow{\Pi} \rho'_L.$$

- Optimal recovery solves $\max_{\mathcal{R}\ \mathrm{CPTP}} F_E(\Pi \circ \mathcal{R} \circ \mathcal{N}_\eta \circ E)$ by SDP (Albert et al., 2017).
- Petz/transpose recovery gives a near-optimal bound without SDP (Zheng et al., 2024).
- We use Petz as the cheaper near-optimal recovery in simulation.

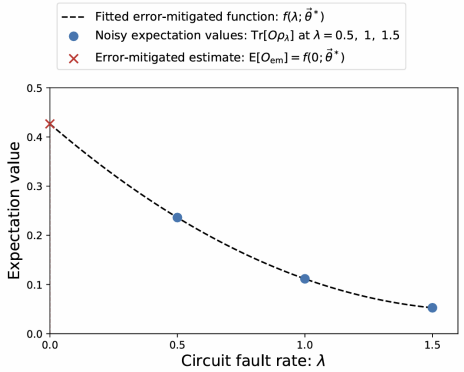


For the Petz choice,

$$N_L := \mathcal{N}_\eta(P_L), \qquad \mathcal{R}(\rho) = P_L \mathcal{N}_\eta^\dagger\left(N_L^{-1/2} \rho N_L^{-1/2}\right) P_L.$$

We evaluate logical observables after this fixed loss, recovery, and readout.

Zero-Noise Extrapolation



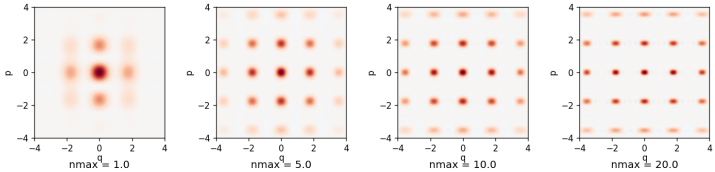
From Z. Cai et al., Rev. Mod. Phys. 95, 045005 (2023)

- QEC applies a recovery map.
- QEM estimates a cleaner observable.
- **Zero Noise Extrapolation (ZNE):** QEM method. Measurement at amplified noise, extrapolate to zero noise limit
- Premise: a smooth, monotonic trend.

GKP code: Seek intrinsic noise knob

Motivation & Outline

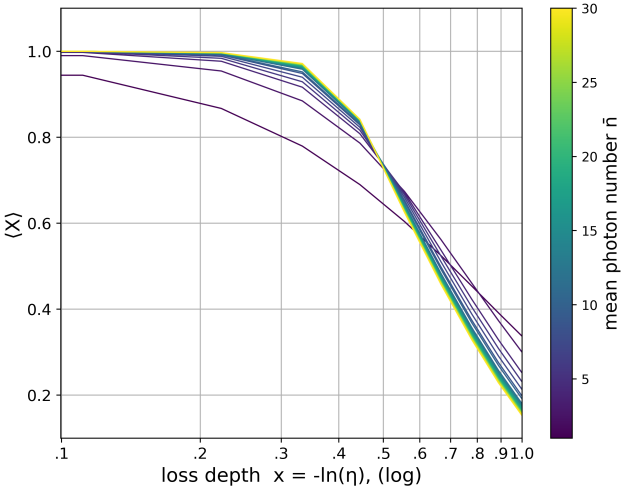
- **Bosonic Code:** Hardware-efficient QEC.
- **Near-term:** Architectures require QEC + QEM.
- **One Practical Bottleneck:** Scaling energy. Experimentally hard.



Wigner Function: Better coherence with increasing energy (mean photon number)

- **This Study:** Exploit code energy ($\bar{n}$) as an intrinsic scaling parameter, s.t.:
  - ZNE under pure loss decoherence
  - Validate on single- and two-qubit observables.
  - Effective resource savings.

Energy as an Intrinsic Scaling Knob

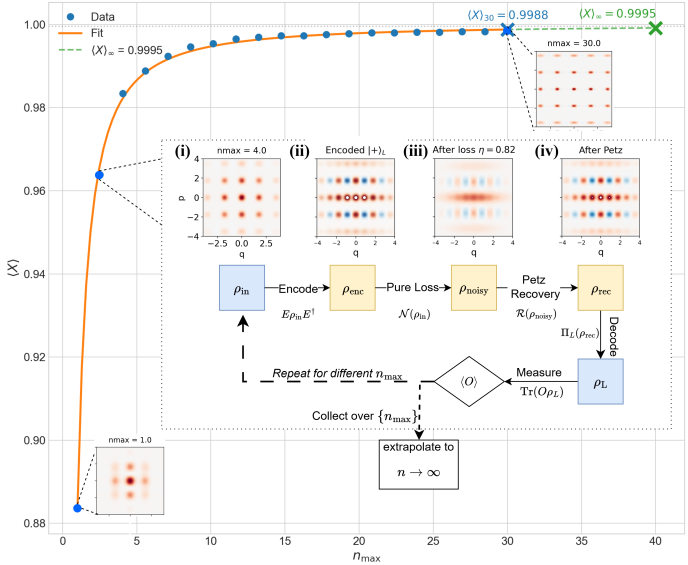


- **Track** $\langle X \rangle$ Under pure loss with near-optimal recovery.

Observe:

- **Correctable Regime ($x \lesssim 0.5$):** Coherence improvement with increasing $\bar{n}$.
- Structured Scaling: good parameter
- **Energy-Scaled Zero-noise Extrapolation (ES- ZNE):** Fix x, vary $\bar{n}$, measure, extrapolate to $\bar{n} \to \infty$.

ES-ZNE Pipeline



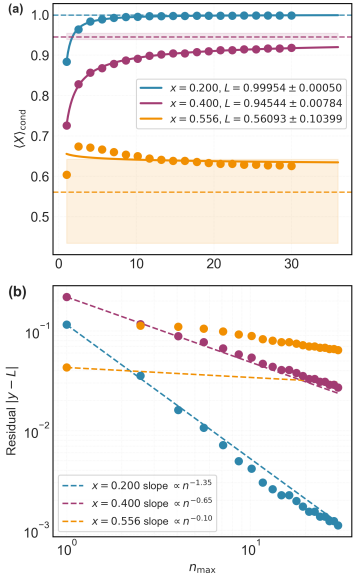
- Construct codeword with **Energy Ladder** at fixed noise level.
- **Logical Observables** from QEC pipeline.
- **Ansatz:** Fit $\{\langle X\rangle\}_n$ to:

$$y(n) = L + c\, n^{-p},$$

where L : **infinite-energy (zero-noise) limit**

- Power-law validated by residual scaling.

Single-Qubit Decoherence Result

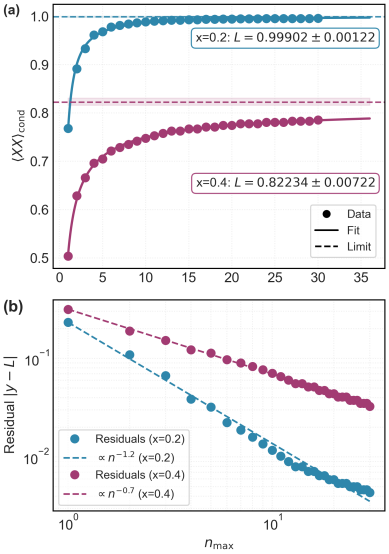


Three Noise Regimes:

- **Shallow Loss ($x = 0.2$):**
  Extrapolation recovers the ideal limit ($L \approx 1$).
- **Threshold ($x = 0.4$):**
  Near pure-loss capacity. Onset of decoherence; still improvement.
- **Decoherence ($x = 0.556$):**
  Beyond capacity; extrapolation breaks down as expected.

Log-Log Residuals (b):
Confirm power-law effective in correctable regime.

Extension to Two Qubits



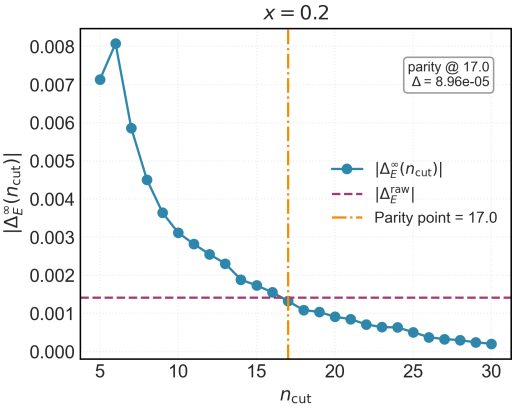
- **State:** Maximally entangled Two Qubit Bell state $|\Phi^+\rangle_L$.
- **Observable:** $\langle XX \rangle$
- **Channels:** Independent pure loss and product recovery maps

Key Result:

Finite-energy trend remains a structured power law; extrapolation successfully mitigates two-qubit errors.

ES-ZNE: General energy scale feature

Practical Payoff: Resource Savings



Compare performance:

- **Benchmark:** Raw coherence at $\bar{n} = 30$
- **Extrapolation:** Extrapolated limit $\Delta_E^{\infty}(n_{\text{cut}})$ with effective energy n_{cut}
- **Parity Point:** For shallow noise, extrapolation matches high-energy benchmark with $\bar{n} \approx 17$.

The QEM Trade-off

Additional sampling time for substantial reduction in required state-prep energy.

Summary and Outlook

- **The Method:** Finite energy limits GKP codes, but gives structured scaling variable. ES-ZNE sweeps $\bar{n}$ at fixed loss and extrapolate to infinite energy limit.
- **The Result:** Recovers ideal, infinite-energy logical expectations in the correctable regime.
- **Resource savings** by trading classical sampling overhead for reduced physical state-preparation demands.
- **Extending the Operating Regime:** Near term low-squeezing hardware $\rightarrow$ future high-squeezing hardware

Next Steps

Applying ES-ZNE to experimental GKP data, where the energy schedule maps directly to the hardware squeezing parameter.

Acknowledgment

Supported by funding from

- the NSF QLCI for Hybrid Quantum Architectures and Networks (NSF award 2016136)
- the University of Wisconsin– Madison L&S Honors Program through The Trewartha Senior Honors Thesis Research Grant.
- Professor Matthew Otten and The Otten Group for mentorship and support.

